Red-light traffic signal with ultrasonic sensor (NET3001 Final Project)

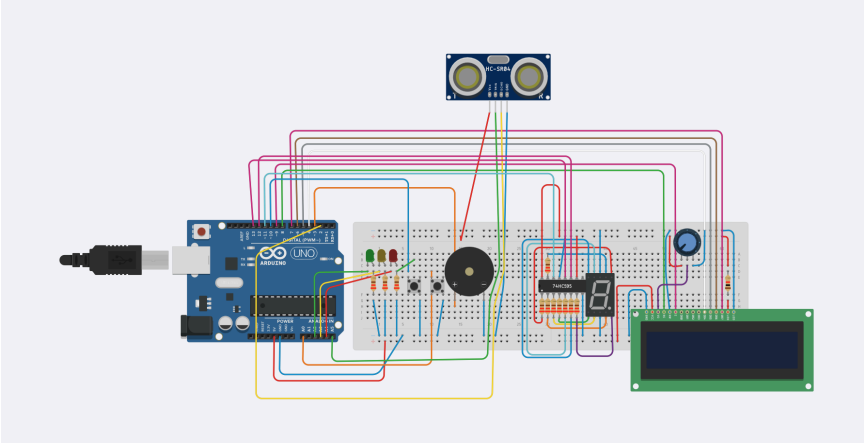
Name of the group members and contributions

- Abdul Rehman
  - TinkerCAD
  - Video
  - serial testing
- Isaac Abdile
  - Hardware
  - Firmware
  - LCD/7-seg
  - report

Brief description of your system

The traffic light works as follows. 10 s green, 5 s amber/yellow, 10 s red. LCD shows the current phase text depending on what state its in. The 7-segment counts down. Buttons pause/resume and reset to green at 10. When red, the ultrasonic detects entering near zone and it will beep once. The LCD will show a violation message(Red light ran).

Circuit diagram



List of components and topics used

Components:

- Arduino Uno
- 3 LEDs
- Resistors
- 2 buttons
- LCD
- 7-seg + 595
- HC-SR04
- Buzzer
- Breadboard

Detailed explanation (pseudo-code)

SETUP
  Configure all GPIO (LEDs, buzzer, ultrasonic trig/echo, shift register, LCD data/control, buttons with pull-ups).
  Start USART at 9600 for text to the serial monitor.
  Initialize LCD (4-bit mode) and 7-segment via shift register.
  Initialize ultrasonic driver and pin-change interrupts for pause and reset.
  Configure Timer1 to interrupt about once per second (used as a 1 s tick, not Arduino delay()).
  Set traffic state: start on GREEN, 10 s countdown on display, normal LCD lines.
  Enable global interrupts (sei) so Timer1 and button ISRs can run.

MAIN LOOP
  Call TrafficLight_step() each iteration.

TrafficLight_step
  PAUSE BUTTON: if a debounced pause click happened, flip running (paused vs running).
  RESET BUTTON: if a debounced reset click happened, go back to GREEN, 10 s, running on, refresh LEDs/segment/LCD and reset ultrasonic “armed” state.

  TIMER
    If a violation message is still showing on the LCD, decrease that counter; when it hits zero, redraw normal phase text.
    Else it should advance the light timer: decrement seconds on 7-seg, or switch phase (green to amber to red to green) when time hits zero.

  RED PHASE and ULTRASONIC (this is only for the case when phase is RED and not paused:
    Read distance in cm. If bad reading, skip.
    If the sensor was “far” before and now reads “near” while armed need to treat as violation:
      beep buzzer, show VIOLATION lines on LCD, print the distance on USART and temporarily disable re-arming.
    If distance is far again re-arm so a new approach can trigger another violation.
    Store current distance for next comparison.

Short reflection

Hardest part:

- tuning zone thresholds and keeping ISRs short.

Next time:

- USART calibration mode; optional Timer1 input capture on echo.